

— ML Performance (2026-05-01 s/d 2026-06-13)

Metric	Value
Evaluation Generated At (UTC)	-
=== 1. MODEL PERFORMANCE SUMMARY ===	
=== 2. RETRAIN HISTORY ===	Periode: 2026-05-01 s/d 2026-06-13
Total Retrain Executions	20
Successful Retrains	20
Retrain Success Rate (%)	100.0
Total Records Trained (cumulative)	15000
Total Feedback Records Used (cumulative)	0
Total Anomalies Found (cumulative)	5000
Total New Patterns Discovered (cumulative)	1
Trigger Type Breakdown	
- manual_upload	20
Recent Retrain Executions (max 10)	
Execution Time Trigger Status Records Feedback Used New Patterns	
2026-06-08 16:16:39 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-06-08 15:24:48 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-06-08 15:03:54 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:37:05 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:56 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:51 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:44 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:36 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:18 manual_upload SUCCESS	records=, feedback=, patterns=0
2026-05-11 14:36:10 manual_upload SUCCESS	records=, feedback=, patterns=0
=== 3. PATTERN DISCOVERY ===	
Top 10 Patterns Category Risk Score Hit Count	
Velocity Fraud Suspect	category=VELOCITY, risk_score=80, hits=150
High Amount Spam Nusa	category=AMOUNT, risk_score=70, hits=80
[Auto Cluster] Potential Fan-In Card Collusion	category=NETWORK, risk_score=60, hits=5
Auto High Amount AGENUSA (5000000+)	category=AMOUNT, risk_score=50, hits=0
AI Discovery: Burst Payment	category=BURST_ATTACK, risk_score=80, hits=0
BURST_AMOUNT	category=BURST, risk_score=50, hits=0
AI Discovery: Brute Force PIN	category=BRUTE_FORCE, risk_score=85, hits=0
AI Discovery: Rapid Retry Declined	category=VELOCITY, risk_score=80, hits=0

Metric	Value
af	category=NETWORK_FAN_IN, risk_score=50, hits=0
Decline Velocity	category=DECLINE_VELOCITY, risk_score=85, hits=0
Total Patterns in System	17
Active Patterns	14
=== 4. FRAUD DETECTION EFFECTIVENESS ===	Periode: 2026-05-01 s/d 2026-06-13
Total New Patterns Generated (all retrains, periode ini)	1
=== CATATAN ===	
Akurasi Model	Tidak tersedia — dataset training bersifat unlabeled (tanpa kolom IS_FRAUD), sehingga metrik Precision, Recall, dan F1-Score tidak dapat dihitung.
Metode Evaluasi	Berbasis threshold boundary dari hasil training (Isolation Forest), bukan validasi terhadap ground truth.
Anomaly Rate vs Fraud Rate	Perbandingan ini bersifat indikatif — anomaly rate adalah parameter model (contamination), sedangkan fraud rate adalah hasil keputusan final (rule engine + pattern + ML + review analisis).